# What the powers of an adjacency matrix count, and how the matrix shows that a graph is connected

A study note for the graph theory course

November 5, 2026

**Abstract**

A graph with $N$ vertices can be written as an $N$ by $N$ table of zeros and ones, its adjacency matrix $A$. This note answers three questions about that table. First, the entry in row $i$ and column $j$ of the $k$-th power of $A$ counts the walks of length $k$ from vertex $n_i$ to vertex $n_j$. It counts walks, not paths. Second, a graph with $N \geq 2$ vertices is connected exactly when every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive, and equivalently when $(I + A)^{N-1}$ has no zero entry. Third, under the course convention a path between two vertices is a walk from one to the other with no repeated vertex. The two theorems are proved, the test is shown on a connected and on a disconnected graph with three vertices, and the test is checked by computation on all graphs with up to 6 vertices.

## 1 Introduction

### 1.1 The three questions and where they came from

This note replies to a student's photographs of a lecture board and to the text of an assignment. Everything on the board that matters is reproduced here, so the reader needs no photograph. The board shows a small drawing with three points labelled 1, 2 and 3, and next to it a table of zeros and ones with the rows 0 1 0, 1 0 1 and 0 1 0 [2, 3]. It says that a graph with $N$ nodes is given through this table, called its adjacency matrix and written $A$, and it asks two questions [2]. A third question comes from the assignment.

**Q1** (board, question 1) What does the entry in row $i$ and column $j$ of the $k$-th power $A^k$ represent?

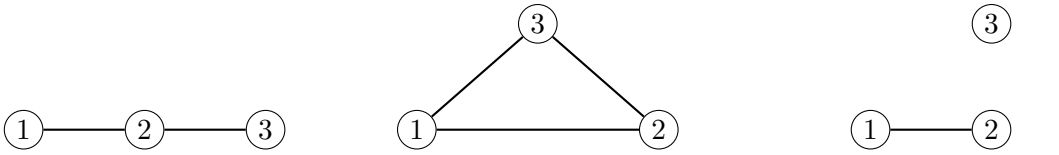
**Q2** (board, question 2) How can one tell from $A$ whether the graph is connected?

**Q3** (assignment) What is the definition of a path between two vertices $n_i$ and $n_j$? The board uses the word *path* in its definition of connected but does not define it [2].

Short answers; every word is defined in Section 2. Q1: the entry counts the walks of length exactly $k$ from $n_i$ to $n_j$ (Theorem 4.4). A walk is a trip along edges that may revisit vertices and reuse edges. The entry counts walks, not paths (Proposition 4.7); a path is a walk that never visits a vertex twice. Q2: the graph is connected exactly when a sum of powers of $A$ has no zero off the diagonal (Theorem 5.6). Q3: a path is a walk that never visits a vertex twice (Definition 2.15). The order of the note follows what each answer needs. The words come first (Section 2), then the comparison of path conventions (Section 3), then Q1 (Section 4) and Q2 (Section 5), and finally the limits (Section 6) and the conclusion (Section 7).

Table 1: The story dictionary: one story word for each technical term.

| Technical term | Story word |
|---|---|
| graph | map |
| vertex | town |
| edge | road |
| adjacent vertices | neighbour towns |
| adjacency matrix | road table |
| walk | trip |
| length of a walk | number of roads on the trip |
| trail | trip that uses no road twice |
| path | trip that visits no town twice |
| cycle | round trip that uses no road twice |
| degree of a vertex | number of roads at a town |
| distance | fewest roads needed between two towns |
| connected graph | every town can be reached from every other |
| component | island of towns |
| identity matrix | stay-at-home table |
| infinite distance | endless |




(a) board map (b) drawing read as three roads (c) one town alone

Figure 1: The three small maps used in this note. Map (a) is the one given by the matrix on the board. Map (b) is the drawing on the board read literally. Map (c) is a map that is not connected.

### 1.2 The story used in the plain-words paragraphs

Before every definition, theorem and worked example there is a short story paragraph, marked in bold at its start. The story is this. A country has towns and roads. Each road joins two different towns. Two towns share at most one road. People make trips along the roads. Figure 1 shows the country of the board. It has three towns in a row. A road joins towns 1 and 2, and a road joins towns 2 and 3. The story paragraphs use only the story words of table 1, which comes right after this paragraph. The story words are only an aid. They never appear in statements or proofs.

### 1.3 Conventions and assumptions

The board and the textbook differ in places, and the board has one mark that cannot be read with certainty. We fix the following conventions. They are cited later by their number.

1. *Finite simple undirected graphs.* Every graph is finite, has no road from a town to itself, no two roads between the same two towns, and no one-way roads. The assignment states this setting, and the symmetric, zero-diagonal matrix on the board fits it [2].

2. *The exponent in Q1 is k.* On the board, question 1 has a small mark above the letter $A$. It is a small letter between quotation-like strokes that could be read as $k$ or $n$ [2, 3]. The assignment, as it was reported, asks about powers $A^k$, so we read $k$. This is an assumption, not something read off the board. It is harmless: the theorem of Section 4 holds for every exponent, whatever it is called.

3. *The matrix, not the drawing, is the example.* The drawing on the board has three lines, between towns 1 and 2 (with an arrowhead at 2), between 1 and 3, and between 2 and 3; fig. 1(b) shows it. The matrix has a 1 only for the pairs $(1, 2)$ and $(2, 3)$ [2, 3]. So the drawing has an extra line between 1 and 3 that the matrix does not have. The problem statement says the graph is given through its matrix, so fig. 1(a) is the example, and the arrowhead is read as a stray mark because the matrix is symmetric. The drawing's three-road map (fig. 1(b)) is also computed, as a side example (Example 5.11); it is not a claim about what the lecturer meant.

4. *Notation.* The board writes $(G, V)$ for a graph, and defines $A_{ij} = 1$ if $e_{ij} \in V$ and $A_{ij} = 0$ if $e_{ij} \notin V$. Here $e_{ij}$ is the edge between $n_i$ and $n_j$. The textbook writes $G = (V, E)$ with $V$ the set of vertices and $E$ the set of edges. So the board's $V$ means the set of edges; the textbook calls that set $E$ [2, 7]. This note always writes $E$, and the board's notation does not appear again. We write $G = (V, E)$, $V = \{n_1, \ldots, n_N\}$, $N = |V|$, and $e_{ij}$ or $n_i n_j$ for the edge joining $n_i$ and $n_j$. The textbook calls the vertices $v_i$; this note uses $n_i$ as on the board. The textbook's page numbers in this note are printed page numbers, which are one less than the page numbers of the PDF file.

5. *The word "path".* Under the course convention, given with the assignment, a path has no repeated vertex. This is the textbook's *simple path*. The textbook's own word "path" means something weaker. Table 2 compares the two.

6. *Walks of length 0.* We allow a walk that is one vertex and no edge. The textbook's definition of a walk does not say so in words, but it assumes it implicitly: its distance properties give $d(u, v) = 0$ exactly when $u = v$ [7, p. 23], and its relation "there is a path between $u$ and $v$" is reflexive [7, p. 49]. We make the convention explicit so that $A^0$, the identity matrix, also counts walks. It is used in Lemma 3.2, in Remark 2.16 and in Appendix A.

7. *The five-town map.* Several examples use the map of fig. 2. It is drawn after the textbook's Fig. 1.2.5 [7, p. 21]. The textbook prints no list of edges, so we read one from the page image. The list is given in the caption of fig. 2. Nothing depends on matching the textbook: the five-town map is defined by that edge list, and the textbook only suggested it.

The notation used throughout is as follows. $A = (A_{ij})$ is the adjacency matrix; $A^k$ is its $k$-th power and $(A^k)_{ij}$ the entry in row $i$ and column $j$ of that power; $I$ is the identity matrix; $\deg(n_i)$ is the degree of $n_i$, and $d(u, v)$, with two arguments, is the distance between $u$ and $v$; $S = A + A^2 + \cdots + A^{N-1}$ is the sum used in the test of Section 5. The letters $N$ (number of vertices), $k$ (length of a walk or exponent), $i, j$ (indices of vertices), $s, t$ (positions or middle indices in sums) and $S$ keep one meaning throughout.

### 1.4 What the sources contain

The textbook [7] defines the adjacency matrix, walks, trails, paths, cycles, distance and connectedness. We cite its numbers only to match names; the proofs in this note do not depend on

it. We searched the textbook's text for the word "matrix" and did not find a theorem about the powers of $A$ or a matrix test for connectivity. This shows only that we did not find them, so we prove both results in this note. Grinberg's lecture notes [1] state the walk-counting theorem for multidigraphs, which are more general than simple graphs, and give the walk-to-path lemma with a length bound. Paramadevan and Sotheeswaran [5] state the connectivity test as their Theorem 2. The details of what was read, and where, are in Appendix B; our comments on their Theorems 1 and 2 are in Section 6.

# 2 Background: graphs, walks and paths

Throughout this section $G = (V, E)$ is a graph as in convention 1, $V = \{n_1, \ldots, n_N\}$, and $u, v, w$ denote vertices. The definitions follow the textbook's order [7]: graph, adjacent, adjacency matrix, degree, walk, closed walk, trail, path, cycle, simple cycle, distance. After each definition come a remark on what it allows and forbids, an example and a non-example. Figure 2 is the map for the examples of walks, trails, paths and cycles.

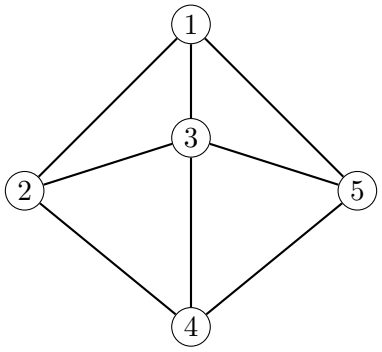



Figure 2: The five-town map used for the examples of this section, drawn after the textbook's Fig. 1.2.5 (convention 7). The label $i$ stands for the vertex $n_i$; these are the textbook's $v_1, \ldots, v_5$. The roads are $n_1n_2$, $n_1n_3$, $n_1n_5$, $n_2n_3$, $n_2n_4$, $n_3n_4$, $n_3n_5$ and $n_4n_5$.

## 2.1 Graphs, adjacency, the adjacency matrix, degree

**In plain words.** A map is a set of towns and a set of roads. Each road joins two different towns. No two roads join the same pair of towns. A road can be driven in both directions.

**Definition 2.1** (simple graph)**.** A *simple undirected graph* $G = (V, E)$ consists of a finite set $V$ of *vertices* and a set $E$ of *edges*, each edge being a set $\{n_i, n_j\}$ of two different vertices, written $n_in_j$ or $e_{ij}$. Hence $n_in_j = n_jn_i$. The number of vertices is $N = |V|$. The textbook uses the notation $G = (V, E)$ [7].

*Remark* 2.2. Because an edge is a set of two different vertices, no edge joins a vertex to itself (no *loop*), and because $E$ is a set, no two edges have the same two ends (no *multiple edges*). Edges have no direction.

*Example.* The board map of fig. 1(a): $V = \{n_1, n_2, n_3\}$ and $E = \{n_1n_2, n_2n_3\}$, so $N = 3$.
*Non-example.* A road from $n_1$ to $n_1$, or two roads between $n_1$ and $n_2$, would not be allowed.

**In plain words.** Two towns are neighbours when one road joins them directly.

**Definition 2.3** (adjacent; textbook Definition 1.1.4)**.** The vertices $n_i$ and $n_j$ are *adjacent* if $n_in_j \in E$ [7, Definition 1.1.4, p. 11].

*Remark* 2.4. Adjacency is a statement about one edge only. It is symmetric, and in a simple graph no vertex is adjacent to itself.

*Example.* In fig. 1(a), $n_1$ and $n_2$ are adjacent.
*Non-example.* In fig. 1(a), $n_1$ and $n_3$ are not adjacent: the road between them exists only in the drawing of fig. 1(b) (convention 3).

**In plain words.** The road table has one row and one column for each town. A one says the two towns are neighbours. A zero says they are not.

**Definition 2.5** (adjacency matrix)**.** The *adjacency matrix* of $G$ is the $N \times N$ matrix $A = (A_{ij})$ with $A_{ij} = 1$ if $n_i$ and $n_j$ are adjacent and $A_{ij} = 0$ otherwise. The textbook defines $A(G)$ in the same way, without a number of its own, after its Definition 1.1.6 [7, p. 11], and the board writes it as $A_{ij}$ [2].

*Remark* 2.6. The matrix records only whether a road exists, never how many; for a simple graph nothing is lost. The textbook notes that $A_{ii} = 0$ for every $i$ and $A_{ij} = A_{ji}$ for all $i, j$ [7, p. 11]: the diagonal is zero because there are no loops, and the matrix is symmetric because edges have no direction.

*Example.* The board matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

is the adjacency matrix of fig. 1(a): $A_{12} = A_{21} = 1$ and $A_{23} = A_{32} = 1$, all other entries 0.
*Non-example.* The matrix with rows 1 1 0, 1 0 0, 0 0 0 is not the adjacency matrix of a simple graph, because $A_{11} = 1$ would be a loop. A matrix with $A_{12} = 1$ and $A_{21} = 0$ is not the adjacency matrix of an undirected graph, because it is not symmetric.

**In plain words.** The number of roads at a town tells how busy the town is.

**Definition 2.7** (degree; textbook Definition 1.2.1)**.** The *degree* $\deg(n_i)$ of a vertex $n_i$ is the number of edges at $n_i$ [7, Definition 1.2.1, p. 16].

*Remark* 2.8. The textbook introduces degree after the adjacency matrix, and this note keeps that order. In a simple graph the edges at $n_i$ correspond one to one to the vertices adjacent to $n_i$, so the degree is also the number of neighbours.

*Example.* In fig. 1(a), $\deg(n_1) = 1$, $\deg(n_2) = 2$ and $\deg(n_3) = 1$.
*Non-example.* The number $N = 3$ of vertices is not a degree: it counts towns, not roads at one town.

## 2.2 Walks, trails, paths and cycles

**In plain words.** A trip goes from a town along a road to the next town, and so on. The trip may come back to a town it already visited. It may use a road again. Count the roads a trip uses. A road used twice is counted twice.

For example, on the board map the list $n_1, n_2, n_3$ is a walk that uses the edges $n_1n_2$ and $n_2n_3$, so its length is 2. The formal definition follows.

**Definition 2.9** (walk; textbook Definition 1.2.10)**.** A *walk of length* $k$ from $u_0$ to $u_k$ is a sequence $u_0, e_1, u_1, \ldots, u_{k-1}, e_k, u_k$ of vertices and edges with $e_s = u_{s-1}u_s$ for $1 \le s \le k$ [7, Definition 1.2.10, p. 21]. Its *length* is $k$.

*Remark* 2.10. Vertices and edges may repeat in a walk, and the length counts every edge each time it occurs [7, p. 22]. In a simple graph the edge between two adjacent vertices is the only one, so a walk is determined by its vertices. The textbook also writes a walk as the sequence of its vertices [7, p. 21], and so does this note: a walk of length $k$ is a list $u_0, \ldots, u_k$ of $k + 1$ vertices in which each vertex is adjacent to the one before it. By convention 6 a single vertex is a walk of length 0.

*Example.* On the five-town map (fig. 2), $n_1, n_3, n_2, n_4, n_3, n_2, n_4, n_3, n_5$ is a walk of length 8. It repeats vertices and edges.
*Non-example.* On the same map, $n_3, n_1, n_4, n_2, n_3, n_5$ is not a walk, because $n_1 n_4$ is not an edge.

**In plain words.** A round trip ends in the town where it began.

**Definition 2.11** (closed walk; textbook Definition 1.2.11)**.** A walk $u_0, \ldots, u_k$ is *closed* if $u_0 = u_k$ [7, Definition 1.2.11, p. 22].

*Remark* 2.12. Nothing else is required: a closed walk may repeat vertices and edges.

*Example.* On the board map (fig. 1(a)), $n_1, n_2, n_1$ is a closed walk of length 2.
*Non-example.* On the same map, $n_1, n_2, n_3$ is a walk but not a closed one.

**In plain words.** Some trips use no road twice. A town may still be visited twice.

**Definition 2.13** (trail; textbook Definition 1.2.12)**.** A *trail* is a walk whose edges $e_1, \ldots, e_k$ are pairwise distinct. The textbook calls this a "path" [7, Definition 1.2.12, p. 22]; "trail" is the English word [6, 1].

*Remark* 2.14. A trail may visit a vertex more than once, but it may not use an edge more than once. To avoid a clash of words, this note writes *trail* for the textbook's "path" of Definition 1.2.12 each time.

*Example.* On the five-town map, $n_1, n_3, n_2, n_4, n_3, n_5$ is a trail. It visits $n_3$ twice but uses each of its five roads once.
*Non-example.* The walk $n_1, n_3, n_2, n_4, n_3, n_2, n_4, n_3, n_5$ is not a trail: it uses the roads $n_2 n_3$, $n_2 n_4$ and $n_3 n_4$ twice each.

**In plain words.** A trip that visits no town twice is the most careful kind of trip.

**Definition 2.15** (path between $n_i$ and $n_j$; textbook Definition 1.2.13, "simple path")**.** Let $n_i, n_j \in V$. A *path between $n_i$ and $n_j$* is a walk $w_0, w_1, \ldots, w_k$ with $w_0 = n_i$, $w_k = n_j$ and $w_0, \ldots, w_k$ pairwise distinct. The number $k$ is its *length*. If $n_i = n_j$, the only path is the one-vertex walk $n_i$, of length 0 (convention 6).

This is the answer to question Q3. For $n_i \neq n_j$ it is the textbook's *simple $(n_i, n_j)$-path*, Definition 1.2.13 [7, p. 22], and it is the English word "path". How these names relate to the textbook's own "path" of Definition 1.2.12 is compared in table 2 of Section 3. The course meaning is given with the assignment (convention 5).

*Remark* 2.16. A path never repeats a vertex, so it never repeats an edge either (Proposition 3.1). A path between a vertex and itself has only one vertex, and so has length 0: by convention 6 it is the sequence $n_i$ alone. A path of length $k$ has $k + 1$ different vertices.

*Example.* On the five-town map, $n_1, n_3, n_4, n_5$ is a path between $n_1$ and $n_5$ of length 3. Another path between the same towns is $n_1, n_5$, of length 1.
*Non-example.* $n_1, n_3, n_2, n_4, n_3, n_5$ is a trail but not a path, because $n_3$ repeats.

**In plain words.** Now look at round trips that use no road twice. Such a trip may still pass through a town more than once.

**Definition 2.17** (cycle; textbook Definition 1.2.14)**.** A *cycle* is a closed trail of length at least 1, which the textbook words as a closed "path" [7, Definition 1.2.14, p. 22].

*Remark* 2.18. A cycle may visit a vertex more than once, but it may not use an edge more than once. The word "cycle" is used here in the textbook's sense, which is broader than a simple cycle (Definition 2.19). A reader who uses "cycle" for a simple cycle should keep the two apart: here a cycle is only a closed trail. Length at least 1 is our reading, so that the one-vertex walk of convention 6 is not a cycle. Cycles are not used again after this section; they are defined to follow the textbook's order.

*Example.* On the five-town map, $n_3, n_2, n_1, n_3, n_5, n_4, n_3$ is a cycle.
*Non-example.* $n_1, n_3, n_2$ is a trail that is not closed, so it is not a cycle.

**In plain words.** Some round trips pass through no town twice before they end. The textbook calls them simple.

**Definition 2.19** (simple cycle; textbook Definition 1.2.15)**.** A cycle is *simple* if only its first and last vertices are equal [7, Definition 1.2.15, p. 22].

*Remark* 2.20. In a simple graph a simple cycle has length at least 3. A cycle of length 1 would need a loop. A cycle of length 2 would be $u, v, u$, which uses the edge $uv$ twice, against the definition of a trail. The textbook writes $C_n$ for the simple cycle on $n$ vertices, with $n \geq 3$ [7, p. 22]. To prevent confusion this note writes "simple cycle" whenever that is meant.

*Example.* On the five-town map, $n_1, n_3, n_4, n_2, n_1$ is a simple cycle of length 4.
*Non-example.* $n_3, n_2, n_1, n_3, n_5, n_4, n_3$ is a cycle but not a simple cycle, because $n_3$ repeats inside it.

**In plain words.** Pick two towns. Look at all trips between them that visit no town twice. The fewest roads on such a trip is the answer. If there is no such trip, we say the answer is endless.

**Definition 2.21** (distance; textbook Definition 1.2.16)**.** The *distance* $d(u, v)$ between $u$ and $v$ is the length of a shortest path from $u$ to $v$ if there is one, and $+\infty$ otherwise.

*Remark* 2.22. The textbook states the definition with its own word "path", that is, with trails [7, Definition 1.2.16, pp. 22–23]. Proposition 3.4 (proved in Section 3) shows that the value is the same with paths, with trails and with walks. A distance is a number or $+\infty$, never a negative number.

*Example.* On the board map, $d(n_1, n_3) = 2$: the only path between them is $n_1, n_2, n_3$.
*Non-example.* On the map of fig. 1(c), $d(n_1, n_3) = +\infty$, which is not a number of roads.

# 3 Question 3: the path between two vertices

The answer to Q3 is Definition 2.15, in its place in the textbook's order: a path between $n_i$ and $n_j$ is a walk from $n_i$ to $n_j$ with no repeated vertex. This section adds only what is new: the comparison with the other meanings of the word, and three facts that the later sections use. The facts are that every path is a trail and every trail is a walk (Proposition 3.1), that every walk contains a path with the same ends and no greater length (Lemma 3.2), and that a path has at most $N - 1$ edges (Proposition 3.4).

## 3.1 Three meanings of one word

The textbook uses the word "path" for a walk with no repeated edge, and needs the extra word "simple" for a walk with no repeated vertex [7, Definitions 1.2.12 and 1.2.13, p. 22]. English texts call the first a trail and the second a path [6, 1]. The course meaning of "path" is the second one (convention 5), which is also the English one. A statement quoted from any source must be translated through table 2 before it is used.

## 3.2 Path, trail, walk

**In plain words.** Every trip that visits no town twice also uses no road twice. A trip that uses no road twice is still a trip. The reverse does not hold. A trip can visit a town twice and use each road once.

Table 2: The three conventions. The last column is the word used in this note.

| What the trip may not do | Textbook word | Textbook number | English word | This note |
|---|---|---|---|---|
| nothing is forbidden | walk | 1.2.10 | walk | walk |
| repeat an edge | "path" | 1.2.12 | trail | trail |
| repeat a vertex | "simple path" | 1.2.13 | path | path |

**Proposition 3.1.** *Every path is a trail and every trail is a walk. Neither statement can be reversed.*

*Proof.* A trail is a walk by Definition 2.13. Let $u_0, \ldots, u_k$ be a path, so its vertices are pairwise distinct, and suppose two of its edges are equal: $e_a = e_b$ with $a < b$. Then $\{u_{a-1}, u_a\} = \{u_{b-1}, u_b\}$, so, since two equal sets of two elements match their elements in one of two ways, there are two cases. If $u_{a-1} = u_{b-1}$, two vertices with different indices $a - 1 \neq b - 1$ are equal. If $u_{a-1} = u_b$, two vertices with different indices $a - 1 \neq b$ are equal. Either way a vertex repeats, which contradicts the distinctness of the vertices. So the edges are distinct and the path is a trail. For the reverse, on the five-town map (fig. 2), $n_1, n_3, n_2, n_4, n_3, n_5$ is a trail that is not a path, and $n_1, n_3, n_2, n_4, n_3, n_2, n_4, n_3, n_5$ is a walk that is not a trail (the examples of Definition 2.13 and Definition 2.15). The textbook gives the same two examples [7, p. 22]. □

### 3.3 A walk contains a path

**In plain words.** Take any trip from one town to another. If it passes a town twice, cut out the loop between the two visits. The trip is now shorter. It still starts and ends where it did. Keep cutting. In the end no town is visited twice. So a trip between two towns can always be cut down. The cut trip visits no town twice. It is never longer than the first.

**Lemma 3.2** (a walk contains a path)**.** *Let $u_0, \ldots, u_k$ be a walk from $u$ to $v$, with $u = v$ allowed and $k \geq 0$. Then there is a path from $u$ to $v$ of length at most $k$, whose vertices are some of $u_0, \ldots, u_k$ in the same order.*

The textbook proves this for $u \neq v$ without a bound on the length (Lemma 1.2.1, [7, p. 23]); Grinberg proves the form with the bound, for $u = v$ as well [1, Corollary 2.9.7, p. 39]. The proof below is ours.

*Proof.* By strong induction on $k$.

*Base case,* $k = 0$. The walk is the single vertex $u = v$, which is a path of length 0.

*Induction hypothesis.* For some $k \geq 1$, the claim holds for every walk of length less than $k$.

*Step.* Let $u_0, \ldots, u_k$ be a walk with $k \geq 1$. If its vertices are pairwise distinct, it is a path of length $k$, and we are done. Otherwise $u_a = u_b$ for some $a < b$. First an example: the walk $1, 2, 3, 2, 4$ has $u_1 = u_3 = 2$, so $a = 1$ and $b = 3$; deleting the 2 vertices $u_2, u_3$ leaves $1, 2, 4$, and the length drops from 4 to 2. In general, delete the $b - a$ vertices $u_{a+1}, \ldots, u_b$; one copy of the repeated vertex, $u_a$, stays. The result is $u_0, \ldots, u_a, u_{b+1}, \ldots, u_k$ (if $b = k$ it ends at $u_a = u_k = v$). Three things hold.

(i) It is still a walk: the only new pair of neighbours is $u_a, u_{b+1}$, and since $u_a = u_b$ and $u_b$ is adjacent to $u_{b+1}$, they are adjacent.

(ii) It has the same ends, and it uses vertices of the original walk in the same order.

(iii) It is shorter: its length is $k - (b - a) < k$.

By the induction hypothesis it contains a path from $u$ to $v$ of length at most $k - (b - a) \leq k$. □

Example: on the five-town map the walk $n_1, n_3, n_2, n_4, n_3, n_2, n_4, n_3, n_5$ of length 8 contains the path $n_1, n_3, n_5$ of length 2. On the board map the closed walk $n_1, n_2, n_1$ contains the path $n_1$ of length 0.

**In plain words.** Suppose you can reach a town by some trip. Then you can reach it by a trip that visits no town twice. The same holds for a trip that uses no road twice. So the kind of trip does not matter. We only ask whether the town can be reached.

**Corollary 3.3** (the three readings agree)**.** *For vertices $u$ and $v$ the following are equivalent: (a) there is a walk from $u$ to $v$; (b) there is a trail from $u$ to $v$; (c) there is a path from $u$ to $v$.*

*Proof.* (c) implies (b) and (b) implies (a) by Proposition 3.1. (a) implies (c) by Lemma 3.2. □

Grinberg states the equivalence of (a) and (c) [1, Proposition 2.9.10, p. 42]; the step through trails, which the textbook's definition of connected needs, is ours. The consequence is that the textbook's definition of connected, which uses its "path", the board's definition, and the reading with paths without repeated vertices describe the same graphs (Section 5). Only the vocabulary changes.

### 3.4 Distance and the longest path

**In plain words.** The fewest roads between two towns is the same whichever kind of trip we use. A trip that visits no town twice can use at most one road less than there are towns.

**Proposition 3.4.** *Let $u, v$ be vertices of a graph with $N$ vertices.*

*(a) If there is a walk from $u$ to $v$, then the shortest walk, the shortest trail and the shortest path from $u$ to $v$ have the same length.*

*(b) Hence $d(u, v) = +\infty$ exactly when there is no walk from $u$ to $v$.*

*(c) Every path has length at most $N - 1$. Hence a finite $d(u, v)$ is at most $N - 1$.*

*Proof.* (a) Let $k$ be the length of a shortest walk. By Lemma 3.2 there is a path of length at most $k$, so of length exactly $k$ by minimality. A path is a trail and a trail is a walk (Proposition 3.1), so the shortest path, trail and walk lengths are all squeezed to $k$. (b) follows from (a) and Corollary 3.3. (c) A path of length $k$ has $k + 1$ pairwise distinct vertices, and $G$ has only $N$ vertices, so $k + 1 \leq N$ [1, Section 2.9.4, p. 40]. □

*Remark* 3.5. By (a), the textbook's distance, defined with its "path" (a trail), has the same value as Definition 2.21.

**In plain words.** Here is the board country, with three towns in a row.

**Example 3.6** (paths on the board map)**.** On the board map (fig. 1(a)) the only path from $n_1$ to $n_3$ is $n_1, n_2, n_3$, of length 2, and the only path from $n_1$ to itself is $n_1$, of length 0. The sequence $n_1, n_2, n_1$ is a walk of length 2 from $n_1$ to $n_1$ but not a path. The distances are $d(n_1, n_2) = 1$, $d(n_1, n_3) = 2$ and $d(n_2, n_3) = 1$. On the map of fig. 1(c) there is no walk from $n_3$ to $n_1$ or to $n_2$, so $d(n_1, n_3) = d(n_2, n_3) = +\infty$ by Proposition 3.4(b).

## 4 Question 1: what the entries of a power of $A$ count

### 4.1 Matrix powers

**In plain words.** To combine the road table with itself, pick a middle town. Ask whether the first town is a neighbour of the middle town. Ask whether the middle town is a neighbour of the last. Do this for every possible middle town and add up the answers.

**Definition 4.1** (matrix product and power)**.** For $N \times N$ matrices $X$ and $Y$, the product $XY$ is the $N \times N$ matrix with

$$(XY)_{ij} = \sum_{t=1}^{N} X_{it} Y_{tj}. \tag{1}$$

The identity matrix $I$ has $I_{ii} = 1$ and $I_{ij} = 0$ for $i \neq j$, and $XI = IX = X$. The powers of $A$ are $A^0 = I$ and $A^{k+1} = A^k A$ for $k \geq 0$.

*Remark* 4.2. Each entry of a product is a sum over a middle index $t$. Since $A$ has only the entries 0 and 1, every entry of every power of $A$ is a non-negative integer. This is the formula whose terms are matched to walks in Theorem 4.4. The exponent $k$ counts how many copies of $A$ are multiplied; $A^0 = I$ is the case of no copies.

*Example.* For the board matrix, $(A^2)_{22} = A_{21}A_{12} + A_{22}A_{22} + A_{23}A_{32} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$.
*Non-example.* $(A^2)_{22}$ is not $A_{22} \cdot A_{22} = 0$. Squaring the matrix is not squaring each entry.

## 4.2 Basic properties of $A$

**In plain words.** The road table has a few simple habits. A town is never its own neighbour. If one town is a neighbour of another, the second is a neighbour of the first. Adding along a row tells how many roads leave that town.

**Proposition 4.3** (properties of $A$ and its powers)**.** *Let $A$ be the adjacency matrix of a simple graph.*

*(a)* $A_{ii} = 0$ *and* $A_{ij} = A_{ji}$ *for all* $i, j$ *[7, p. 11].*

*(b) Every power of $A$ is symmetric:* $(A^k)_{ij} = (A^k)_{ji}$.

*(c) The $i$-th row sum of $A$ is the degree:* $\sum_j A_{ij} = \deg(n_i)$. *The column sums are equal to the row sums.*

*(d)* $(A^2)_{ii} = \deg(n_i)$.

The *transpose* $X^T$ of a matrix $X$ has $(X^T)_{ij} = X_{ji}$; it swaps rows and columns, and a matrix with $X^T = X$ is symmetric. The rule $(XY)^T = Y^T X^T$ is used below.

Part (d) is used in Proposition 4.7; parts (b) and (c) are given for completeness and are not used later.

*Proof.* (a) is the textbook's remark after the definition of $A(G)$, restated in Definition 2.5.

(b) By (a), $A^T = A$. Using $(XY)^T = Y^T X^T$ repeatedly, $(A^k)^T = (A^T)^k = A^k$.

(c) The sum $\sum_j A_{ij}$ counts the vertices adjacent to $n_i$. In a simple graph these correspond one to one to the edges at $n_i$, so the sum is $\deg(n_i)$ by Definition 2.7. By (a) the column sums are the same.

(d) By Eq. (1) with $X = Y = A$ and by (a), $(A^2)_{ii} = \sum_t A_{it}A_{ti} = \sum_t A_{it}^2$. Since $A_{it} \in \{0, 1\}$ we have $A_{it}^2 = A_{it}$, so $(A^2)_{ii} = \sum_t A_{it} = \deg(n_i)$ by (c). □

Paramadevan and Sotheeswaran state (d) in the proof of their Theorem 1 [5, p. 17]. The textbook states (a) and does not state (b) to (d), so those are ours. On the board matrix the row sums are $1, 2, 1$, which are the degrees of the example of Definition 2.7.

## 4.3 The walk-counting theorem

**In plain words.** Multiply the road table by itself a few times. Each number in the result tells how many trips of exactly that many roads lead from one town to another. Trips may come back to a town they visited. Trips may use a road again. Every such trip is counted.

**Theorem 4.4** (walk counting)**.** *Let $G$ be a simple undirected graph with vertices $n_1, \dots, n_N$ and adjacency matrix $A$. For every integer $k \geq 0$ and all $i, j \in \{1, \dots, N\}$, the entry $(A^k)_{ij}$ equals the number of walks of length $k$ from $n_i$ to $n_j$.*

Grinberg proves the corresponding theorem for multidigraphs, which includes $k = 0$ [1, Theorem 4.5.10, pp. 120–123]. The textbook has no such theorem (Section 1.4). The proof below is ours and is written for simple graphs.

*Proof.* Write $W_k(i, j)$ for the number of walks of length $k$ from $n_i$ to $n_j$. By Remark 2.10, $W_k(i, j)$ is the number of sequences $u_0, \dots, u_k$ of vertices with $u_0 = n_i$, $u_k = n_j$ and $u_{s-1}$ adjacent to $u_s$ for $1 \leq s \leq k$. We prove $W_k(i, j) = (A^k)_{ij}$ by induction on $k$.

*Base case, $k = 0$.* The only walk of length 0 that starts at $n_i$ is the single vertex $n_i$. So $W_0(i, j) = 1$ if $i = j$ and 0 otherwise, which is $I_{ij} = (A^0)_{ij}$. (For $k = 1$ directly: a walk of length 1 from $n_i$ to $n_j$ is one edge, so $W_1(i, j) = A_{ij}$.)

*Induction hypothesis.* For some $k \geq 0$, $W_k(i, t) = (A^k)_{it}$ for all $i, t$.

*Step.* Take a walk $u_0, \dots, u_{k+1}$ of length $k + 1$ from $n_i$ to $n_j$, and let $n_t = u_k$ be its next-to-last vertex. Dropping the last vertex leaves a walk of length $k$ from $n_i$ to $n_t$, and $n_t$ is adjacent to $n_j$. Conversely, a walk of length $k$ from $n_i$ to $n_t$, followed by $n_j$ when $n_t$ is adjacent to $n_j$, is a walk of length $k + 1$ from $n_i$ to $n_j$. Different $t$ give different walks, because $t$ can be read off the walk. Hence

$$W_{k+1}(i, j) = \sum_{t=1}^{N} W_k(i, t)\, A_{tj},$$

because $A_{tj} = 1$ allows exactly one last step from $n_t$ to $n_j$ and $A_{tj} = 0$ allows none. By the induction hypothesis the right side is $\sum_t (A^k)_{it} A_{tj} = (A^{k+1})_{ij}$ by the definition of the product, Eq. (1), and $A^{k+1} = A^k A$ (Definition 4.1). This closes the induction. □

*Remark* 4.5. The theorem counts walks of exactly length $k$, and it needs no assumption that the graph is connected. A zero entry $(A^k)_{ij} = 0$ says only that no walk of length exactly $k$ joins the two vertices; a walk of another length may exist. For $k = 2$ and $i = j$ the theorem restates Proposition 4.3(d): $(A^2)_{ii}$ counts the closed walks $n_i, n_t, n_i$, one for each neighbour $n_t$.

## 4.4 Worked example on the board matrix

**In plain words.** Here is the board country with three towns in a row. We multiply the road table by itself and read a few numbers. Each number counts trips of the chosen number of roads.

**Example 4.6** (the board matrix)**.** For the board matrix $A$ of Definition 2.5,

$$A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad A^3 = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}.$$

The entry $(A^2)_{13} = 1$ counts the walk $n_1, n_2, n_3$. The entry $(A^2)_{22} = 2$ counts the walks $n_2, n_1, n_2$ and $n_2, n_3, n_2$. The entry $(A^3)_{12} = 2$ counts $n_1, n_2, n_1, n_2$ and $n_1, n_2, n_3, n_2$. The diagonal of $A^2$ is $1, 2, 1$, the degrees. The entry $(A^2)_{12} = 0$, although $n_1$ and $n_2$ are adjacent, because no walk of length exactly 2 joins them: the map has no triangle. Exact length is what matters.

For the drawing of fig. 1(b) read as three roads (convention 3) the matrices are

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{pmatrix}.$$

Here $(A^3)_{11} = 2$ counts the closed walks $n_1, n_2, n_3, n_1$ and $n_1, n_3, n_2, n_1$. These matrices are small enough to check by hand, for example $(A^3)_{12} = \sum_t (A^2)_{1t} A_{t2} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$ for the board matrix. They were also computed by exact integer arithmetic and the walks were listed by direct enumeration (Appendix B).

### 4.5 Walks are not paths

**In plain words.** The table counts trips. A trip may go to a neighbour town and come straight back. Such a trip visits a town twice, so it is not a trip that visits no town twice. The table can therefore show a number where no such trip exists.

**Proposition 4.7** (walks, not paths)**.** *The entry $(A^k)_{ij}$ counts walks, and in general neither paths nor trails. In particular, if $n_i$ has a neighbour, then $(A^2)_{ii} = \deg(n_i) > 0$ although there is no path of length 2 from $n_i$ to $n_i$.*

*Proof.* A path between $n_i$ and itself has pairwise distinct vertices with first and last equal, so it has one vertex and length 0 (Definition 2.15). By Proposition 4.3(d), $(A^2)_{ii} = \deg(n_i)$, which is positive when $n_i$ has a neighbour. The walks it counts are $n_i, n_t, n_i$, and each of these uses the edge $n_i n_t$ twice, so none is a trail. On the board map, $(A^3)_{12} = 2$ (Example 4.6). A graph with 3 vertices has no path of length 3 by Proposition 3.4(c), so these entries do not count paths. The two walks counted are $n_1, n_2, n_1, n_2$, which uses the edge $n_1 n_2$ three times, and $n_1, n_2, n_3, n_2$, which uses the edge $n_2 n_3$ twice; so neither is a trail. □

Table 3: On the five-town map (fig. 2), the numbers of walks, trails and paths of length $k$ from $n_1$ to $n_5$, and the entry $(A^k)_{15}$. The last column equals the walk count in every row. The counts were obtained by listing all walks of each length on the map with the program `appendices/computations.py` (Appendix B).

| $k$ | walks | trails | paths | $(A^k)_{15}$ |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 |
| 2 | 1 | 1 | 1 | 1 |
| 3 | 8 | 3 | 3 | 8 |
| 4 | 16 | 7 | 3 | 16 |
| 5 | 72 | 6 | 0 | 72 |
| 6 | 192 | 6 | 0 | 192 |
| 7 | 704 | 16 | 0 | 704 |
| 8 | 2112 | 0 | 0 | 2112 |

Table 3 shows the difference on a larger map: the matrix entry follows the walk column and no other. The path column is 0 from $k = 5$ on because a path has at most 4 edges here (Proposition 3.4(c)). A trail uses each of the 8 roads at most once, so it has at most 8 edges. The count 0 for trails of length 8 has a reason. Such a trail would use every road exactly once. In a trail that uses every road, each vertex other than the two ends is entered and left equally often, so all its roads come in pairs and its degree is even. Hence at most two vertices, the ends, can have odd degree. On this map the degrees $\deg(n_1), \ldots, \deg(n_5)$ are $3, 3, 4, 3, 3$, so four vertices have odd degree, and no such trail exists.

What can be said about paths is this. If $i \neq j$ and $(A^k)_{ij} \geq 1$, then there is a walk of length $k$ (Theorem 4.4), hence a path between $n_i$ and $n_j$ of length at most $k$ (Lemma 3.2). The value of the entry does not give the number of paths.

# 5 Question 2: deciding connectivity from $A$

## 5.1 Connected graphs and components

**In plain words.** In some countries you can drive from any town to any other town. In others, some town is cut off from the rest.

**Definition 5.1** (connected; textbook Definition 2.1.1)**.** A graph $G$ is *connected* if for any two vertices $u$ and $v$ there is a path between $u$ and $v$ [7, Definition 2.1.1, p. 48]. The board states the same definition: a graph is called connected if for any $n_i, n_j$ there exists a path between the nodes [2].

*Remark* 5.2. The textbook's word "path" in Definition 2.1.1 is its trail, while the board and the course use the path of Definition 2.15. Corollary 3.3 shows that this does not change which graphs are connected: a walk, a trail and a path between two vertices exist for the same pairs. A graph with one vertex is connected, since there are no two different vertices to join.

*Example.* The board map of fig. 1(a) is connected: the paths $n_1, n_2$ and $n_2, n_3$ and $n_1, n_2, n_3$ join all pairs of distinct vertices, and every vertex is joined to itself by a path of length 0.
*Non-example.* The map of fig. 1(c) is not connected: $n_3$ has no neighbour, so no walk and hence no path joins $n_3$ to $n_1$ (Corollary 3.3).

**In plain words.** Say two towns are linked when one can reach the other. Every town is linked to itself. If the first is linked to the second, the second is linked to the first. If the first is linked to the second and the second to a third, the first is linked to the third. So the towns split into groups of towns that are all linked to one another.

**Lemma 5.3** (reachability is an equivalence relation)**.** *The relation "there is a path between $u$ and $v$" on $V$ is reflexive, symmetric and transitive.*

*Proof.* Reflexive: a vertex is joined to itself by a path of length 0. Symmetric: a path read backwards is a path. Transitive: a walk from $u$ to $v$ followed by a walk from $v$ to $w$ is a walk from $u$ to $w$, which contains a path by Lemma 3.2. □

The textbook states that this relation is an equivalence relation [7, p. 49]. An *equivalence class* is a group of vertices that can all reach one another.

**In plain words.** Towns that can reach each other form an island. Towns on different islands cannot reach each other. A country with one island is one where every town can be reached from every other.

**Definition 5.4** (connectivity component; textbook, p. 49)**.** The classes of the relation of Lemma 5.3 induce subgraphs of $G$, called the *connectivity components* of $G$ [7, p. 49]. The subgraph induced by a set $X$ of vertices has the vertices in $X$ and the edges of $G$ with both ends in $X$.

*Remark* 5.5. A graph is connected if and only if it has exactly one component ([7, Remark 2.1.1, p. 49]; also [1, Definition 2.9.12, p. 42]). Two vertices lie in the same component exactly when a path, and by Corollary 3.3 a walk, joins them.

*Example.* The map of fig. 1(c) has two components: the one with vertices $n_1, n_2$ and the road $n_1n_2$, and the one with the single vertex $n_3$.
*Non-example.* $\{n_1, n_3\}$ is not the vertex set of a component of the map of fig. 1(c), since no path joins $n_1$ and $n_3$.

## 5.2 The test

**In plain words.** To see whether one town can reach another, look at trips of one road, of two roads, and so on. Add up the tables for all these trips. A zero in a place means no such trip joins the two towns, whatever the number of roads. We only need to try trips of up to one road less than there are towns. A trip that visits no town twice never has more roads than that.

**Theorem 5.6** (connectivity test)**.** *Let $G$ be a simple graph with $N \geq 2$ vertices and adjacency matrix $A$, and put $S = A + A^2 + \cdots + A^{N-1}$. Then $G$ is connected if and only if $S_{ij} > 0$ for every $i \neq j$.*

The textbook has no matrix test (Section 1.4). Paramadevan and Sotheeswaran state this test as their Theorem 2 [5, p. 17]. The proof below is ours and uses Theorem 4.4, Lemma 3.2 and Proposition 3.4(c).

*Proof. Connected implies $S_{ij} > 0$.* Let $G$ be connected and let $i \neq j$. By Definition 5.1 there is a path between $n_i$ and $n_j$ (by Corollary 3.3, whichever of the readings of "path" is used); let $k$ be its length. Since $i \neq j$, $k \geq 1$, and by Proposition 3.4(c) $k \leq N - 1$. A path is a walk, so by Theorem 4.4 $(A^k)_{ij} \geq 1$. Every power of $A$ has non-negative entries (Remark 4.2), so $S_{ij} \geq (A^k)_{ij} \geq 1$.

*$S_{ij} > 0$ for all $i \neq j$ implies connected.* Fix $i \neq j$. The number $S_{ij}$ is the sum of the non-negative integers $(A^k)_{ij}$ for $k = 1, \ldots, N-1$, so one of them is positive: some $k$ with $1 \leq k \leq N-1$ has $(A^k)_{ij} \geq 1$. By Theorem 4.4 there is a walk of length $k$ from $n_i$ to $n_j$, and by Lemma 3.2 a path between $n_i$ and $n_j$. A vertex is joined to itself by the path of length 0. So any two vertices are joined by a path, and $G$ is connected. □

*Remark* 5.7. The step from a positive entry to a path is where care is needed. The entry counts walks (Proposition 4.7), and a walk need not be a path. Lemma 3.2 supplies the step, and Proposition 3.4(c) shows that powers up to $N - 1$ are enough. In our reading, the published proof of the same test relies on its Theorem 1 counting paths and so does not supply this step (Section 6). For $N = 1$ the sum $S$ is empty and the single-vertex graph is connected, so the test is stated for $N \geq 2$.

**In plain words.** There is a neater way to write the same test. Add a stay-at-home option to the road table. Then multiply that table by itself a number of times, one less than the number of towns. The test then reads: no zero anywhere.

**Corollary 5.8** (the form with $(I+A)^{N-1}$)**.** *A simple graph $G$ with $N \geq 1$ vertices is connected if and only if the matrix $(I + A)^{N-1}$ has no zero entry.*

The idea is this. The binomial theorem holds for matrices $X, Y$ that commute, that is, $XY = YX$, and $I$ commutes with every matrix, so $(I + A)^{N-1}$ expands into $\sum_{k=0}^{N-1} \binom{N-1}{k} A^k$, a sum with positive coefficients of the same powers as in $S$ plus $I$. The exponent $N - 1$ is the same bound as in $S$: it is the longest length a path can have. The diagonal is positive because of $I$, and an off-diagonal entry is positive exactly when the same entry of $S$ is. The full proof, including the binomial theorem for commuting matrices, is in Appendix A; Example 5.9 checks the case $(I + A)^2 = I + 2A + A^2$. No source that we read states the test in this form with a proof; the result is derived in this note.

## 5.3 Worked examples

**In plain words.** First the board country, with three towns in a row. We list the road table, the table for trips of two roads, and their sum. Then we look for zeros.

**Example 5.9** (a connected map)**.** The board map has $N = 3$, so $S = A + A^2$ and the other form is $(I + A)^2$. The matrices are

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

$$I + A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad (I + A)^2 = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{pmatrix}.$$

Every off-diagonal entry of $S$ is positive and $(I + A)^2$ has no zero entry, so by Theorem 5.6 and Corollary 5.8 the map is connected, as it should be. The entry $S_{13} = 1$ comes from $(A^2)_{13} = 1$, the walk $n_1, n_2, n_3$. The entry $(A^2)_{12} = 0$ although $n_1$ and $n_2$ are adjacent; the entry $A_{12} = 1$ makes up for it in $S$. This is why the powers are summed and a single power is not enough. As a check of Corollary 5.8, $(I + A)^2 = I + 2A + A^2$ here.

**In plain words.** Now a country where one town has no road. We do the same calculation and find zeros.

**Example 5.10** (a map that is not connected)**.** Take the map of fig. 1(c), with the single road $n_1n_2$ and $n_3$ alone. Then

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$I + A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad (I + A)^2 = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The matrix $S$ has zeros at $(1, 3)$, $(2, 3)$ and their mirror images, and $(I + A)^2$ has zeros in the same places. By Theorem 5.6 and Corollary 5.8 the map is not connected. By Corollary 5.12, stated below, the components are $\{n_1, n_2\}$ and $\{n_3\}$.

**In plain words.** For the drawing on the board, read with a third road, every pair of towns is joined by a road. We expect no zeros.

**Example 5.11** (the drawing read as three roads)**.** For the map of fig. 1(b),

$$S = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}, \qquad (I + A)^2 = \begin{pmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}.$$

Every entry of $S$ is 2, including the diagonal, and every entry of $(I + A)^2$ is 3. The map is connected, which is also visible directly.

### 5.4 Components, distance and sharpness

**In plain words.** The table of sums also tells which towns share an island. A zero in a place means the two towns are on different islands.

**Corollary 5.12** (components from $S$)**.** *For $i \neq j$, the vertices $n_i$ and $n_j$ lie in the same component of $G$ if and only if $S_{ij} > 0$. So the pairs with $S_{ij} = 0$ are exactly the pairs in different components.*

*Proof.* If $n_i$ and $n_j$ are in the same component, a path joins them (Remark 5.5), and the first half of the proof of Theorem 5.6, applied to this pair alone, gives $S_{ij} > 0$. If $S_{ij} > 0$, the second half of that proof gives a path between $n_i$ and $n_j$, so they are in the same class of the relation of Definition 5.4. □

*Remark* 5.13 (distance). For distinct vertices in the same component, $d(n_i, n_j)$ is the least $k \geq 1$ with $(A^k)_{ij} > 0$. The statement is not new: Paramadevan and Sotheeswaran give it as their Theorem 3 [5, p. 17], and as item 7 of their Section 2.2, which counts walks [5, p. 16]. Their proof of Theorem 3 reads a nonzero entry as a path; the proof here, which goes through Lemma 3.2, is ours. Indeed, a walk of length $k$ exists exactly when the entry is positive (Theorem 4.4), and the shortest walk has the length of the shortest path (Proposition 3.4(a)). On the board map, $A_{13} = 0$ and $(A^2)_{13} = 1$, so $d(n_1, n_3) = 2$, in agreement with Example 3.6.

**In plain words.** Take towns in a row, each joined only to the next. To get from the first town to the last, you must pass every road in between. So no shorter trip exists. If we only looked at shorter trips, we would wrongly say the first and last towns are cut off.

**Proposition 5.14** (the limit $N-1$ cannot be lowered)**.** *Let $N \geq 2$ and let $G$ be the path graph with vertices $n_1, \ldots, n_N$ and edges $n_i n_{i+1}$ for $1 \leq i < N$. Then $(A^k)_{1N} = 0$ for $k < N-1$ and $(A^{N-1})_{1N} = 1$. Hence a sum $A + \cdots + A^m$ with $m < N-1$ has a zero at position $(1, N)$, although $G$ is connected.*

*Proof.* Each edge of $G$ joins two vertices whose indices differ by exactly 1, so each step of a walk changes the index by $+1$ or $-1$. A walk from $n_1$ to $n_N$ must change the index from 1 to $N$, a total change of $N-1$, so it has at least $N-1$ steps. By Theorem 4.4 this gives $(A^k)_{1N} = 0$ for $k < N-1$. A walk of length $N-1$ has $N-1$ steps of $\pm 1$ with sum $N-1$, so every step is $+1$; hence the only one is $n_1, n_2, \ldots, n_N$, and $(A^{N-1})_{1N} = 1$. □

A computer run on the path graphs with $N = 3, 4, 5, 6$ gave the first positive power $A^{N-1}$ each time, as the proposition says [4].

## 5.5 Breadth-first search, for comparison

**In plain words.** Stand in town 1. Visit all its neighbour towns. Then visit the neighbours of those. Note each town once. If every town was visited, then every town can be reached from town 1.

This section is a comparison and a cross-check; the note does not rely on it. *Breadth-first search* starts with the set $\{n_1\}$ and repeatedly adds every $n_j$ with $A_{mj} = 1$ for some $n_m$ already in the set. When nothing can be added, $G$ is connected exactly when the set contains all $N$ vertices. The vertices added are those joined to $n_1$ by a path, so by Remark 5.5 the set is the component of $n_1$. Grinberg states that in a breadth-first search tree the tree path from the root is a shortest path [1, p. 176]; using the search to decide connectivity is our reading, not a statement of that source. By our own count, forming $S$ takes $N-2$ matrix products: $A^2, \ldots, A^{N-1}$, each from the previous one by one product. The search reads the row of each visited vertex at most once. Matrix products and row reads cannot be compared directly, so we do not compare their cost. On the maps of Example 5.9 and Example 5.10 the search visits $n_1, n_2, n_3$ and $n_1, n_2$ respectively [4].

## 5.6 A computer check

This is an experimental finding, not a proof. A *labelled* graph has its vertices fixed as $n_1, \ldots, n_N$, and each of the $N(N-1)/2$ pairs of vertices either is an edge or is not, so there are $2^{N(N-1)/2}$ labelled simple graphs. A program listed all of them for $N = 1, 2, \ldots, 6$ (1, 2, 8, 64, 1024 and 32768 graphs, of which 1, 1, 4, 38, 728 and 26704 are connected; we have not compared these

counts with an outside source). For each graph it decided connectivity by the sum test, by the $(I + A)^{N-1}$ test and by breadth-first search from $n_1$ (the previous subsection), and the three answers always agreed [4]. The proofs above do not rest on this check.

# 6 Discussion: limits and open points

**What the results do not say.** The entry $(A^k)_{ij}$ gives the number of walks, not of paths (Proposition 4.7), and for connectivity only whether it is positive matters. The size of an entry is not a measure of how well two vertices are joined: table 3 shows entries growing to 2112 at $k = 8$ while no path on this five-town map has more than 4 edges (Proposition 3.4(c)). The test of Theorem 5.6 decides connectivity; it does not list the paths. Nothing in this note claims what the lecturer intended beyond what the board and the assignment say.

**Scope of the proofs.** Both main results are stated and proved for finite simple undirected graphs (convention 1). The proof of Theorem 4.4 identifies a walk with its vertex sequence and an edge with the entry 1, and we do not claim the result for graphs with loops or multiple edges. For directed graphs the matrix is not symmetric and "connected" splits into more than one notion, so the test is not claimed there; Grinberg's version of the walk-counting theorem covers directed graphs [1, Theorem 4.5.10].

**Two statements in a published paper.** Paramadevan and Sotheeswaran [5, p. 16] write: "the number of different paths of length $l$ between the $i$th and $j$th vertices of $G$ is equal to the $(i, j)$th entry of $A_G^l$" (their Theorem 1). The only point we make is about the word "path". With "path" meaning no repeated vertex, the statement is not correct as written, by Proposition 4.7; with "path" meaning "walk" its content is our Theorem 4.4. Their own Section 2.2, item 7, states the count with "walks" [5, p. 16], so the word "paths" in their Theorem 1 conflicts with their own preliminaries. Their Theorem 2 is the test of Theorem 5.6. The last line of its proof reads: "Conversely, if $b_{ij} \neq 0$, then there exist at least one path from $v_i$ to $v_j$, thus $G$ is connected" [5, p. 17]. In our reading this passes from a nonzero entry of a sum of powers to a path without the step of Lemma 3.2, which turns a walk into a path (Remark 5.7); this is an interpretation of their text, and we use their paper for the statement of the test only.

**Strength of the computer checks.** The exhaustive comparison up to $N = 6$ (Section 5) and the path graphs of Proposition 5.14 cover finitely many cases. The first supports the proofs without replacing them; the second only illustrates a proposition that is proved.

# 7 Conclusion

**Q1.** For a finite simple undirected graph with adjacency matrix $A$, the entry $(A^k)_{ij}$ is the number of walks of length $k$ from $n_i$ to $n_j$ (Theorem 4.4), for every $k \geq 0$. Walks may repeat vertices and edges, so the entry does not count paths or trails (Proposition 4.7). For example, on the board map $(A^2)_{11} = 1$ counts the walk $n_1, n_2, n_1$. This is the textbook's walk, Definition 1.2.10; the textbook has no such theorem, so the proof is given here.

**Q2.** A graph with $N \geq 2$ vertices is connected if and only if every off-diagonal entry of $S = A + A^2 + \cdots + A^{N-1}$ is positive (Theorem 5.6), and equivalently if and only if $(I + A)^{N-1}$ has no zero entry (Corollary 5.8). The proof uses the walk-counting theorem, the lemma that a walk contains a path of no greater length (Lemma 3.2), and the fact that a path has at most $N - 1$ edges (Proposition 3.4(c)). On the board map every off-diagonal entry of $S$ is 1, so the map is connected; on the map with an isolated town $S$ has zeros, so it is not (Example 5.9, Example 5.10).

**Q3.** A path between $n_i$ and $n_j$ is a walk $w_0, \ldots, w_k$ from $n_i$ to $n_j$ whose vertices are pairwise distinct (Definition 2.15). This is the textbook's simple path, Definition 1.2.13. The textbook's own word "path", Definition 1.2.12, means a walk with no repeated edge, which this note calls a trail. By Corollary 3.3 the choice among walk, trail and path does not change which graphs are connected.

## References


[1] Darij Grinberg. *An introduction to graph theory.* arXiv:2308.04512v3 [math.HO], Spring 2025 edition dated June 7, 2025; text for Math 530, Drexel University. 2025. URL: `https://arxiv.org/abs/2308.04512` (visited on Nov. 5, 2026).

[2] *Lecture whiteboard, photo 1.* Local file sources/whiteboard-1.jpg. Photo supplied with the assignment.

[3] *Lecture whiteboard, photo 2.* Local file sources/whiteboard-2.jpg. Photo supplied with the assignment; same board as photo 1 from another angle.

[4] *m3_connectivity.py: matrices, tests and exhaustive check.* Local file scripts/m3_connectivity.py. Written for this paper; output in scripts/m3_connectivity.out.

[5] P. Paramadevan and S. Sotheeswaran. "Properties of Adjacency Matrix of a Graph and It's Construction". In: *Journal of Science, Eastern University, Sri Lanka (JSc EUSL)* 12.1 (2021), pp. 13–21. URL: `https://www.fsc.esn.ac.lk/jsc/archive/12.1.2.pdf` (visited on Nov. 5, 2026).

[6] *Path (graph theory).* Wikipedia, English edition, section "Walk, trail, and path". URL: `https://en.wikipedia.org/wiki/Path_(graph_theory)` (visited on Nov. 5, 2026).

[7] P. A. Petrosyan, V. V. Mkrtchyan, and R. R. Kamalyan. *Grafneri tesutyun (Graph Theory), teaching manual, in Armenian.* Local file; supplied with the assignment. Page numbers are printed page numbers, one less than the PDF page numbers. The title is transliterated from Armenian script. Yerevan: Yerevan State University Press, 2015. ISBN: 978-5-8084-1995-7.


## A Derivation of the form with $(I + A)^{N-1}$

This appendix proves Corollary 5.8: a simple graph $G$ with $N \geq 1$ vertices is connected if and only if $(I+A)^{N-1}$ has no zero entry. It uses Theorem 5.6 and the notation of Definition 4.1 and Remark 4.2.

*Proof of Corollary 5.8.* Two matrices $X, Y$ commute if $XY = YX$; the identity matrix commutes with every matrix, because $IX = XI = X$. For this reason the binomial expansion holds for $I + A$, and we prove it in two lines. For every $m \geq 0$, $(I + A)^m = \sum_{k=0}^{m} \binom{m}{k} A^k$. For $m = 0$ both sides are $I$. If it holds for $m$, then

$$(I + A)^{m+1} = (I + A)^m (I + A) = \sum_{k=0}^{m} \binom{m}{k} A^k + \sum_{k=0}^{m} \binom{m}{k} A^{k+1} = \sum_{k=0}^{m+1} \left[ \binom{m}{k} + \binom{m}{k-1} \right] A^k,$$

using $IA^k = A^k$ and Pascal's rule $\binom{m}{k} + \binom{m}{k-1} = \binom{m+1}{k}$ (with $\binom{m}{-1} = \binom{m}{m+1} = 0$). With $m = N - 1$ this gives

$$(I + A)^{N-1} = \sum_{k=0}^{N-1} \binom{N-1}{k} A^k, \tag{2}$$

where $\binom{N-1}{k}$ is the number of ways to choose $k$ items from $N - 1$ items, and is at least 1 for $0 \leq k \leq N - 1$. The exponent $N - 1$ is the same bound as in $S$: it is the longest length a

path can have (Proposition 3.4(c)). By Remark 4.2 every $A^k$ has non-negative integer entries, so every term of (2) is non-negative.

*Diagonal entries.* The term with $k = 0$ is $\binom{N-1}{0} I = I$, which contributes 1 to every diagonal entry. All other terms are non-negative, so every diagonal entry of $(I + A)^{N-1}$ is at least 1 for every graph.

*Off-diagonal entries.* For $i \neq j$ the term $k = 0$ contributes 0, so

$$\big((I + A)^{N-1}\big)_{ij} = \sum_{k=1}^{N-1} \binom{N-1}{k} (A^k)_{ij}.$$

The coefficients are positive and the terms $(A^k)_{ij}$ are non-negative, so this sum is positive exactly when at least one $(A^k)_{ij}$ with $1 \leq k \leq N - 1$ is positive. That is exactly when $S_{ij} > 0$, because $S_{ij}$ is the sum of the same non-negative terms.

*Conclusion.* Hence $(I + A)^{N-1}$ has no zero entry if and only if $S_{ij} > 0$ for all $i \neq j$. For $N \geq 2$ this is connectivity by Theorem 5.6. For $N = 1$ the matrix is $I = (1)$, which has no zero entry, and a graph with one vertex is connected (Definition 5.1). □

On the board map, (2) reads $(I + A)^2 = I + 2A + A^2$, which was checked by exact integer arithmetic (Example 5.9). No source that we read states or proves the test in this form, so this derivation is the only support for it.

# B Sources, checks and how to rerun the computations

**Sources and what was read.** The *course textbook* [7] is a local file supplied with the assignment. Its title is in Armenian script; the reference list gives a transliteration because the document is typeset with a Latin font. We viewed the page images of printed pages 11, 21, 22, 23, 48 and 49 before quoting definitions, and read the extracted text of pages 16 and 22–23. The rest of the book was searched, not read: a search of the extracted text for the word "matrix" finds the adjacency matrix only on printed pages 7, 11, 12, 34 and 61, and finds no theorem about the powers of $A$ and no matrix test for connectivity. The extraction drops most symbols, so this does not prove that no such statement exists. The two *board photographs* [2, 3] show the same board from two angles and were read together. *Grinberg's lecture notes* [1] were read in Section 2.9, in Theorem 4.5.10 and its proof, and in the passage on breadth-first search. *Paramadevan and Sotheeswaran* [5] were read in Section 3.1 (Theorems 1 to 3 and their proofs). The *encyclopedia page* [6] was read as raw wikitext on 2026-11-05 and is used only for names. The *computer check* [4] is a short program written for this note; it computes the examples of Section 5 and the exhaustive comparison. The program `appendices/computations.py` described below repeats those computations and adds the others, so every number in the text can be rerun from that one program.

**What is ours.** The statements and proofs of Lemma 3.2, Corollary 3.3, Proposition 3.4, Proposition 4.3(b)–(d), Theorem 4.4, Theorem 5.6, Corollary 5.8, Corollary 5.12, and Proposition 5.14 are proved in this note. Where a source states the same result, the text says which one. The reading of the board is recorded as conventions 1 to 3 in Section 1.3.

**Rerunning the numbers.** Every matrix, count and example in the text is recomputed, with exact integers, by the program `appendices/computations.py` in the paper folder. From that folder, run `python3 -I appendices/computations.py`; it takes about twenty seconds, and its output is saved as `appendices/computations.out.txt`. The program lists the matrices of Example 4.6, Example 5.9, Example 5.10 and Example 5.11, the walks that these examples name, the statuses (walk, trail, path, closed) of the examples in Section 2, the counts of table 3,

a check of the binomial identity, a consistency check of the cutting rule of Lemma 3.2 on all short walks of the five-town map, and the exhaustive comparison with breadth-first search for $N \leq 6$. These checks support the proofs and do not replace them.